

GEN 03042

Molecular analysis of the *pac* gene encoding a puromycin *N*-acetyl transferase from *Streptomyces alboniger*

(Recombinant DNA; nucleotide sequence homologies; promoter activity; transcription initiation)

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Received by C.R. Hutchinson: 17 January 1989

Revised: 28 February 1989

Accepted: 1 March 1989

SUMMARY

Nucleotide sequence of a 906-bp fragment of *Streptomyces alboniger* DNA containing the gene (*pac*), which encodes a puromycin *N*-acetyltransferase (PAC), has been determined. The *pac* gene contains a 600-nt open reading frame, starting with an ATG codon, which encodes a polypeptide of M_r 21 531; this is consistent with the 23 ± 1.5 kDa size of the PAC enzyme. High-resolution S1 mapping indicates that transcription starts at or next to a C residue 35 bp upstream from the putative ATG start codon. A 263-bp DNA fragment from the 5' region of the *pac* gene has promoter activity in the promoter-probe plasmid pIJ486. Its –35 and –10 regions show significant structural homology to the corresponding regions of the *hyg* gene promoter, but they are different from the promoter sequences of other *Streptomyces* and *Escherichia coli* genes.

INTRODUCTION

In numerous actinomycetes the biosynthesis of antibacterial antibiotics is highly regulated and generally commences at the end of the log phase of growth (Berdy, 1980). The presence of a resistance mecha-

nism is required to protect the producing organism from the toxic effect of the antibiotic. Self resistance may be achieved by a variety of means, including enzymatic drug inactivation (Cundliffe, 1984). The expression of the gene(s) for resistance appears to take place coordinately with the expression of the

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Abbreviations: *amp*, gene encoding β -lactamase; Ap, ampicillin; APH, aminoglycoside phosphotransferase; *aph*, gene encoding APH; bp, base pair(s); DMPM, *O*-demethyl puromycin

O-methyltransferase; *dmpM*, gene encoding DMPM; fd, phage fd transcription terminator; *hyg*, gene encoding hygromycin phosphotransferase; Km, kanamycin; *mel*, tyrosinase-encoding gene; nt, nucleotide(s); ORF, open reading frame; PAC, puromycin *N*-acetyltransferase; *pac*, gene encoding PAC; Pr, *pac* promoter; RBS, ribosome-binding site; Ts, thiostrepton; *tsp*, transcription start point; *tsr*, gene encoding Ts resistance; u, unit(s).

genes for biosynthesis. Moreover, there is biochemical evidence suggesting that the modifying enzymes act on the earliest toxic intermediate (Vara et al., 1985b; Pérez-González et al., 1985; Murakami et al., 1986).

As a model system we are studying the molecular genetics and biochemistry of puromycin biosynthesis by *S. alboniger*. Two genes, *pac* and *dmpM*, encoding PAC and DMPM, respectively, have already been cloned (Vara et al., 1985a; 1988). The PAC and DMPM activities are involved in puromycin resistance and catalysis of the synthesis of one of the last intermediates of the pathway, respectively (Vara et al., 1985b; Pérez-González et al., 1985). The gene *dmpM* is tightly linked to and downstream from *pac* in the genome of *S. alboniger*, which suggests that other genes of the pathway may be linked to them (Vara et al., 1988). In addition, *pac* and *dmpM* appear to be expressed concomitantly during the log phase of growth of *S. alboniger*, suggesting that their transcription is coordinately regulated (Vara et al., 1985b). Examination of the *pac* promoter and *pac* transcription is, therefore, important to understand the mechanism(s) controlling puromycin biosynthesis. In addition, the gene *pac* has been cloned

and expressed in higher cells (Vara et al., 1986). Plasmid constructs containing *pac*, which use puromycin as the selection agent, have proven to be useful as cloning vectors for a variety of mammalian cells (de la Luna et al., 1988).

In this work we have determined the nucleotide sequence of *pac* and have identified its promoter and *tsp*.

EXPERIMENTAL AND DISCUSSION

(a) Sequencing of the *pac* gene

The restriction map and sequencing strategy of the *pac*-encoding 906-bp *Sal*I-*Nco*I DNA fragment from *S. alboniger*, contained in plasmid pVN3.1, are presented in Fig. 1. The nucleotide sequence is outlined in Fig. 2. The direction of transcription of *pac* was previously known (Fig. 1; Vara et al., 1985a) and, therefore, ORFs were examined in the relevant DNA strand. Only two ORFs starting with ATG (nt 254) and GTG (nt 275), and both terminating by TGA (nt 851) could correspond to *pac*.

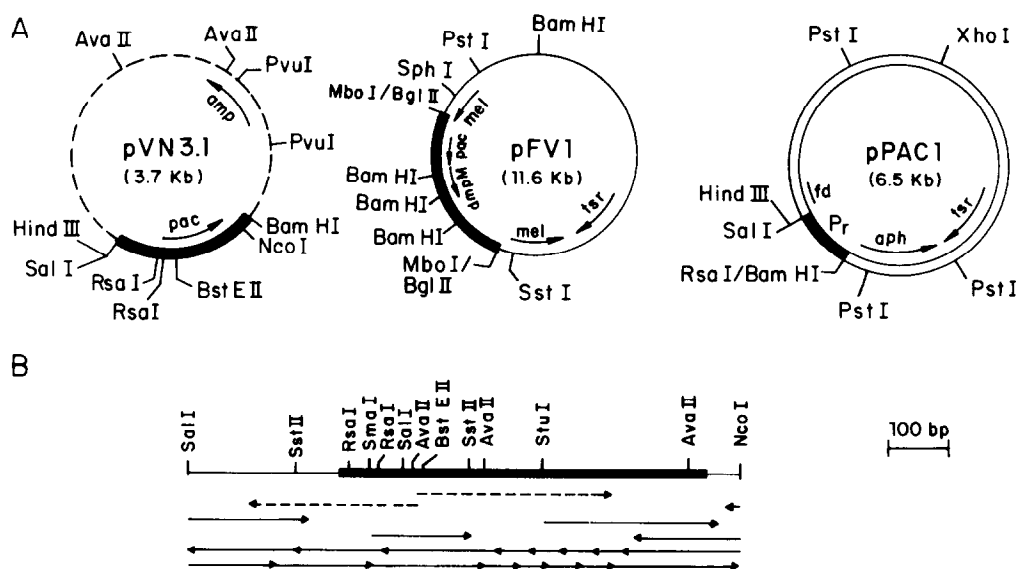


Fig. 1. Restriction maps of several plasmids and of the 906-bp fragment containing *pac*. (A) Restriction maps of plasmids pVN3.1 and pFV1 were based on those published by Vara et al. (1985a; 1988) and that of pPAC1, a derivative of the promoter-probe plasmid pIJ486 (Ward et al., 1986), was obtained in this work. DNA from pUC19 (dashed line) (Yanisch-Perron et al., 1985); *S. alboniger* (thick line); pIJ702 (thin line) and pIJ486 (open box). Arrows indicate the direction of transcription. (B) Sequencing strategy. Discontinuous and continuous lines indicate chemical (Maxam and Gilbert, 1980) and 'dideoxy' enzymatic (Hattori and Sakaki, 1986) sequencing procedure, respectively. The arrows indicate the direction and length of sequence determined. The heavy line represents the coding region of the *pac* gene.

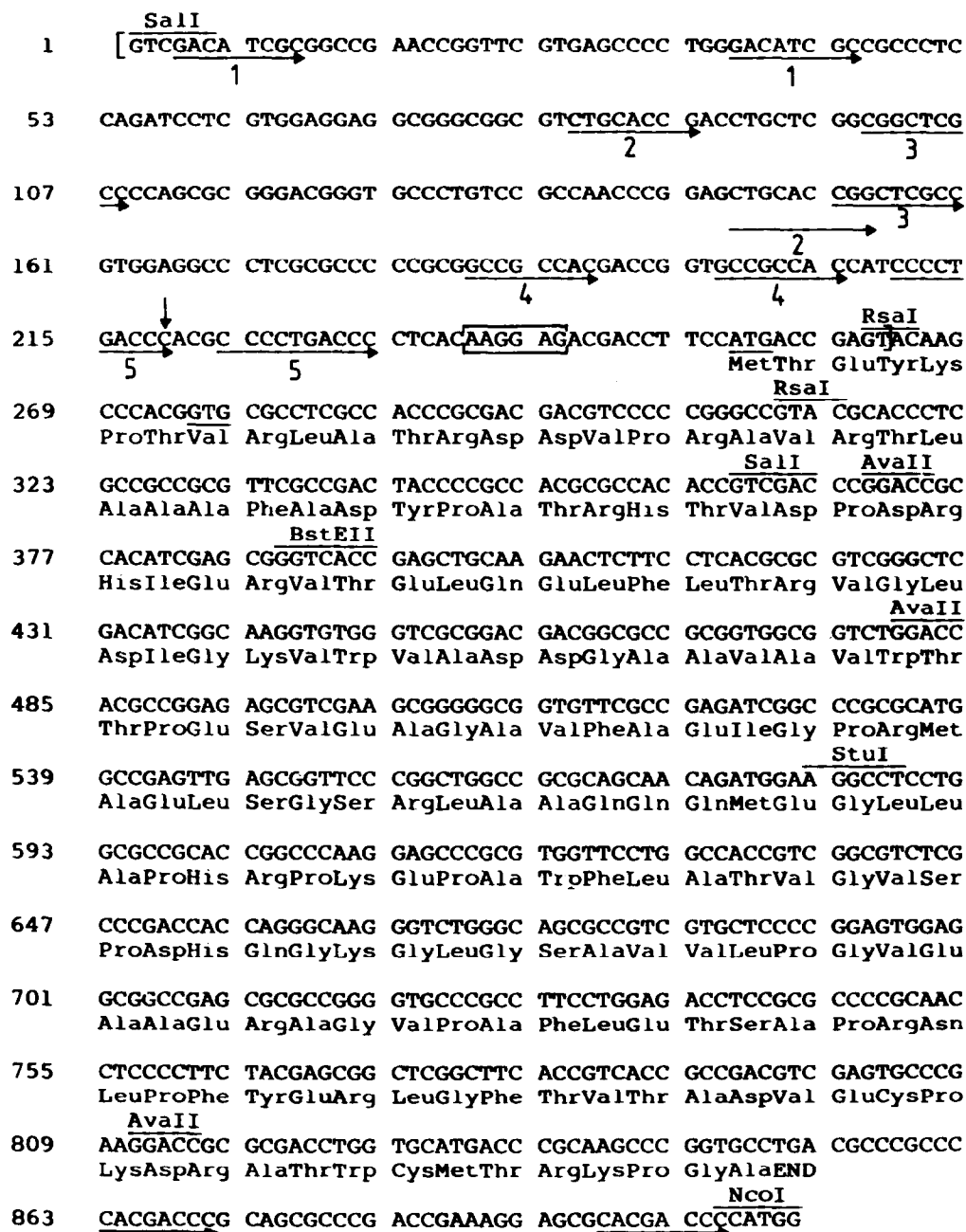


Fig. 2. Nucleotide sequence of the 906-bp *pac* fragment (see Fig. 1B). The ATG and GTG potential start codons are underlined. Horizontal arrows indicate direct repeat sequences. The vertical arrow indicates the transcription start point. The potential RBS is boxed. Brackets delimit the 263 bp fragment used in the cloning of the *pac* promoter. *E. coli* strain JM83 (Vieira and Messing, 1982) and plasmid pUC19 (Yanisch-Perron et al., 1985), used for sequencing purposes, were obtained from J.E. Davies. *E. coli* was grown using either liquid or solid LB medium supplemented with 40 µg XGal/ml and 100 µg Ap/ml as required. *E. coli* was transformed as described by Maniatis et al. (1982).

The deduced amino acid sequences indicated polypeptides of 21531 Da (ORF1) and 20663 Da (ORF2), both close to that (23 ± 1.5 kDa) determined for PAC by physicochemical methods (Vara

et al., 1985b). However, ORF1 seems to be the correct one because a putative RBS (AAGGAG) is optimally located 10 bp before its start codon; this RBS appears to be too distant (31 bp) from the GTG



Fig. 3. Determination of the *tsp* of the *pac* gene. RNA from *S. lividans* harboring plasmid pFV1, was prepared as described by Zalacain et al. (1986). Plasmid pVN3.1 was prepared as described by Zalacain et al. (1986). Plasmid pVN3.1 was digested with *Bst*EII, 5' end-labeled with [γ - 32 P]ATP and then cut with *Hind*III. The resulting 406-bp DNA fragment (50 ng) was isolated (Vara et al., 1985a) and then hybridized to 100 μ g RNA from a *S. lividans* clone harboring pFV1. S1 mapping was performed as described by Favaro et al. (1980) except that DNA-RNA hybrids were formed at 60°C, because of the high G + C content of *Streptomyces* DNA. After S1 (150 u) digestion, the

start codon of ORF2 (Fig. 2). The % of G + C in the first, second and third position of the ORF1 codons is 73.0, 52.5 and 94.5%, respectively, which is in accordance with the rules of Bibb et al. (1984) for *Streptomyces* genes. The codon usage for *pac* (not shown) is very similar to that found for other *Streptomyces* genes (Zalacain et al., 1986; Thompson and Gray, 1983; Bibb et al., 1985; Epp et al., 1987; Horinouchi et al., 1987; Tohyama et al., 1987).

The deduced amino acid sequence for *pac* shows no significant homology with those deduced from the genes for other acetyl transferases (not shown). Several direct repeats are located 5' to the *pac* coding sequence (Fig. 2). Different direct repeats have been found in similar regions of other *Streptomyces* genes. Although their significance, if any, is unknown, it has been suggested that they might be involved in interaction with DNA-binding proteins (Bibb et al., 1984; Tohyama et al., 1987).

(b) Transcription studies

A high-resolution S1 mapping experiment was performed to determine the transcription start point for the *pac* gene. The results (Fig. 3) show that the *tsp* lies at or next to the C, 35 bp upstream from the putative ATG start codon (Fig. 2). Because of the high G + C content of the *S. alboniger* DNA fragment used to construct the RNA/DNA hybrid relative to that of the pUC19 used as a sequence ladder, it is conceivable that the actual *tsp* of the *pac* gene could differ slightly from the one experimentally found, although we have no evidence to that effect. Nevertheless, this finding is compatible with the contention that ORF1 is the correct coding sequence for PAC (see above).

(c) Isolation of the *pac* promoter

A 282-bp *Hind*III-*Rsa*I DNA fragment containing the upstream region of *pac* was isolated from

reaction products were loaded on a 6% polyacrylamide sequencing gel in parallel with sequencing reactions (Hattori and Sakaki, 1986) from a fragment of known sequence from plasmid pUC19. Lanes labeled G, A, T and C contain standards from sequencing reactions. Numbers on the left indicate the size (in nt) of the standards and those on the right the length (in nt) of the DNA fragment(s) protected from the nuclease S1 digestion.

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